

The empirical recurrence for OEIS A260245, recovered from its tail

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Abstract

OEIS A260245 records “Empirical recurrence of order 86” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 134$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 90$. Its last coefficient is $c_{86} = -400 \neq 0$, so no further tail satisfies anything shorter; any order-86 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A260245 is “Number of $(n+2) \times (5+2)$ 0..1 arrays with each 3×3 subblock having clockwise perimeter pattern 00000001 00000101 or 00001001.”. It has offset 1 and begins

1598, 3026, 8242, 14045, 38280, 122048, 238852, 523742, ...

The entry states:

Empirical recurrence of order 86 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 12:03:51, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 134$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 134$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 86: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 86.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 268$ exact terms from $n = 4$ on, it returns order 86 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{86} c_i a(n-i),$$

c_1	0	c_{23}	142150	c_{45}	−20937038	c_{67}	−1783844
c_2	0	c_{24}	3029329	c_{46}	−12601256	c_{68}	1364588
c_3	26	c_{25}	5082976	c_{47}	26151314	c_{69}	−446340
c_4	39	c_{26}	4020770	c_{48}	4114123	c_{70}	243996
c_5	0	c_{27}	−1709334	c_{49}	9923152	c_{71}	−413972
c_6	−276	c_{28}	−8514065	c_{50}	−30031497	c_{72}	224192
c_7	−784	c_{29}	−12412902	c_{51}	−10167702	c_{73}	−72200
c_8	−618	c_{30}	−8426931	c_{52}	−2820746	c_{74}	−3312
c_9	1608	c_{31}	3953018	c_{53}	21557830	c_{75}	39808
c_{10}	6515	c_{32}	15703715	c_{54}	37768289	c_{76}	−60964
c_{11}	9946	c_{33}	24727196	c_{55}	11681528	c_{77}	18392
c_{12}	−454	c_{34}	13789785	c_{56}	9117080	c_{78}	−25860
c_{13}	−31274	c_{35}	−1907048	c_{57}	−22510726	c_{79}	16352
c_{14}	−69530	c_{36}	−25387117	c_{58}	−12861867	c_{80}	496
c_{15}	−59208	c_{37}	−33523120	c_{59}	−5131408	c_{81}	7376
c_{16}	69743	c_{38}	−18273946	c_{60}	−8207048	c_{82}	2448
c_{17}	283250	c_{39}	−1722382	c_{61}	6844908	c_{83}	1312
c_{18}	417947	c_{40}	34892480	c_{62}	758348	c_{84}	−880
c_{19}	138980	c_{41}	25063194	c_{63}	3042116	c_{85}	0
c_{20}	−645053	c_{42}	26200287	c_{64}	178515	c_{86}	−400
c_{21}	−1502872	c_{43}	−12384458	c_{65}	−2139164		
c_{22}	−1533220	c_{44}	−25308639	c_{66}	2961828		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 90$, and it is the recurrence OEIS A260245 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{86} - c_1x^{86-1} - \dots - c_{86}$. Its constant term is $-c_{86} = 400$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 86 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 86, so $\deg q = 86$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 86, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 23 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 86, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -400 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-86 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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